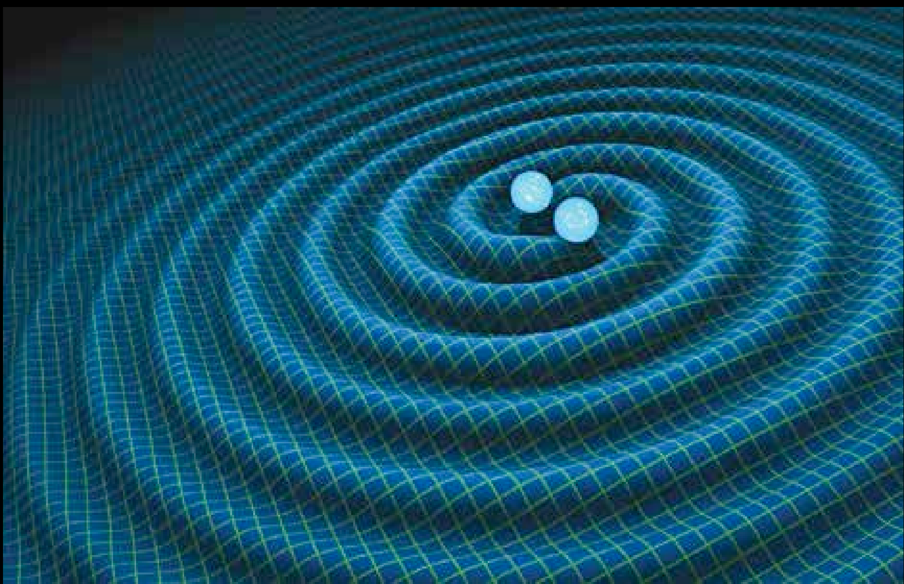
mathematical ideologies). Space and time are now an interwoven fabric, think of it as a giant membrane that exists everywhere (and every...when?) around us. So far, we have no objects or any physics at all! It's just one giant, happy, undisturbed membrane. For the sake of simplicity, picture a 2 dimensional membrane, which corresponds to one space dimension and one time dimension. Now, drop a tennis ball onto this membrane in your head. The membrane, which was stretched out now sinks under the weight of the ball. According to Einstein's theory of relativity, this is precisely what happens to spacetime in the presence of mass! The presence of any object warps space and time.



Ripples or waves in the interwoven fabric of space and time

So how does this explain gravity? Go back to the membrane you have in your head, and place a coin somewhere close to the ball. The coin will automatically start rolling down to the centre because of the dent produced by the ball and, if you give the coin the right amount of momentum in the right direction, the coin will start spinning around the tennis ball. This can also be verified at home using a simple bowl: any coin with the right amount of velocity will start spinning around the centre of the bowl. Of course, eventually the real world experiment will fall to the centre because friction causes the velocity of the coin to slow down, but this sort of nuisance doesn't bother cosmic objects. The exact mechanism that is explained above is why planets continue to revolve around the sun, it's all due to the warp in spacetime that the sun produces.

This is a very handy idea. To understand gravity, we simply consider the warps produced by massive objects. This is why physicists are obsessed with geometry, everything we need to know about the masses and the dynamics of the body can now be understood, at a fundamental level, simply by the curvature the object produces. It stands to wit that this curvature produced by a single object will move with the object. This was the idea that prompted Einstein to predict the existence of gravitational waves. An object that is moving will produce a moving gravitational field; a series of ripples as the curvature produced by the body moves with it, like waves produced when moving your hand through a pond of water. Since these ripples are curvatures in spacetime, we should be able to observe it as a 'springy' effect, a continuous contraction and expansion effect as the ripples move past us. An experimental detection of gravitational waves will, once and for all, conclusively prove Einstein's theory of gravity.

This is precisely why the Laser Interferometer Gravitational Wave Observatory (LIGO) was set up in the United States of America. The experimental data not only validates relativity but also provides a novel method of studying astronomical objects that are at a great distance from us simply by considering the gravitational waves they produce. The experimental setup took a mammoth 8 years to construct, and was completed in the year 2002. There are two facilities which together comprise the LIGO experiment: The LIGO Livingstone observatory in Livingston, Louisiana and the DOE Hanford site in Richland, Washington. Each observatory has an L shaped ultra-vacuum system measuring nearly 4 kilometres in length, where the experiments are carried out.

These detectors, despite the enormous complexity involved in construction and effective noise filtration, have a very simple principle at their very core: the interference of light. It works by having two long mirrors placed perpendicular to each other, a fair distance apart. The mirrors are freely suspended and are therefore allowed to move. A laser light, after it has been optically filtered to produce exactly one particular frequency is split into two beams and each beam is reflected off one mirror (the actual experiment uses a device called a Fabry-Perot cavity to make it reflect back and forth multiple times, but that is merely to improve the quality of the signal) and recombined with the other beam. Now, if the path that the light travels is the same for both beams, then the two light beams will have traversed an equal number of wavelengths and will therefore interfere constructively, i.e., every crest of one beam will correspond to a crest of the other beam and similarly for every trough. However, if a gravitational wave were to pass through the experimental facility, we would expect one of the two perpendicular paths to shift its total length. Therefore the two waves will no longer constructively interfere. By measuring how they interfere, scientists can then calculate how much the path length changed and for how long; correspondingly they can calculate the strength of the wave and how long the signal lasted for.

The problem with detecting gravitational waves is that these fields are weak. Very weak. So weak in fact, that even random signals generated by the sound of airplanes soaring many thousands of meters overhead are enough to disturb